

ALENCON-VALFRAMBERT AP, France

WMO: 071390

Lat: 48.4456N

Lon: 0.1103E

Elev: 463

StdP: 14.45

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	26.2	17.6	13.6	27.2	21.4	16.3	28.1	25.7	49.0	23.4	49.0	4.9	30	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	86.1	67.3	81.9	65.9	78.4	64.7	69.2	81.8	67.5	78.5	65.9	75.8	7.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	94.4	73.6	63.5	89.3	71.8	62.1	84.7	70.1	33.6	82.2	32.1	78.8	30.9	75.9	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.3	16.2		DB	17.9	92.1	5.1	4.6	14.2	95.4	11.2	98.1	8.4	100.7	4.7	104.0
				WB	17.2	72.4	5.0	2.2	13.6	73.9	10.7	75.2	7.9	76.4	4.2	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	40.1	41.2	45.4	49.7	55.7	61.8	65.6	65.0	59.3	53.5	45.8	40.9
	DBStd	10.80	7.07	6.33	5.69	5.73	5.58	5.52	5.31	5.19	4.94	5.75	6.02	6.88
	HDD50	1274	310	248	162	73	12	0	0	0	1	31	149	288
	HDD65	4905	772	666	608	458	291	127	58	64	183	356	575	747
	CDD50	2031	3	2	20	65	190	354	482	465	279	141	24	6
	CDD65	185	0	0	0	0	4	31	75	64	11	0	0	0
	CDH74	1970	0	0	0	10	67	354	741	643	148	7	0	0
	CDH80	583	0	0	0	0	5	100	227	220	30	1	0	0
Wind	WSAvg	7.5	8.3	8.5	8.3	7.8	7.5	7.2	7.3	6.7	6.7	6.8	7.5	8.1
Precipitation	PrecAvg	29.5	3.0	2.4	2.1	2.1	2.4	2.0	2.2	2.0	2.2	2.9	2.9	3.3
	PrecMax	40.2	6.3	5.3	6.3	5.1	3.9	5.2	5.6	5.2	5.6	6.8	6.2	7.3
	PrecMin	20.9	0.4	0.3	0.2	0.4	0.4	0.2	0.3	0.2	0.5	0.9	0.9	0.8
	PrecStd	5.8	1.5	1.4	1.3	1.3	1.1	1.2	1.4	1.4	1.5	1.5	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	58.7	67.4	75.4	80.1	88.6	91.0	92.8	84.0	73.6	62.1	56.6
		MCWB	53.2	50.9	54.4	59.5	65.1	70.4	69.3	69.8	65.4	62.3	58.1	54.2
	2%	DB	53.8	54.3	62.2	70.1	75.5	82.4	85.5	85.9	78.5	68.5	58.7	54.7
		MCWB	51.7	50.0	52.5	57.6	62.5	66.8	66.9	67.2	63.3	60.6	55.5	52.6
	5%	DB	51.7	52.2	58.2	65.1	71.4	77.8	81.8	80.7	73.6	64.8	56.6	53.0
		MCWB	49.7	49.0	51.4	54.9	60.5	64.8	65.9	65.5	61.3	58.7	53.9	51.1
	10%	DB	49.7	50.4	54.9	60.5	67.3	73.8	77.7	76.4	69.7	62.3	54.6	51.0
		MCWB	47.7	47.7	49.9	52.5	58.2	62.9	64.7	64.0	60.1	57.9	52.2	49.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	52.9	56.3	62.0	66.8	71.9	71.1	71.8	67.9	64.6	59.1	55.0
		MCDB	55.1	55.8	62.8	72.3	77.2	84.7	85.6	88.0	79.1	70.6	60.5	56.0
	2%	WB	52.1	51.1	54.1	58.4	64.1	68.4	68.9	69.0	65.2	61.9	56.2	53.0
		MCDB	53.6	53.4	59.2	67.5	73.4	78.8	81.7	81.5	74.5	66.4	58.2	54.2
	5%	WB	50.0	49.5	52.5	55.8	61.3	66.0	67.2	67.1	63.2	60.2	54.2	51.3
		MCDB	51.5	51.8	56.7	63.3	69.1	75.8	78.6	77.7	71.2	63.7	56.2	52.7
	10%	WB	47.9	47.8	50.7	53.4	59.1	63.9	65.6	65.1	61.4	58.5	52.4	49.4
		MCDB	49.4	50.0	54.6	59.5	66.1	72.0	75.4	74.1	67.4	61.6	54.4	50.9
Mean Daily Temperature Range	5% DB	MDBR	9.3	11.7	15.1	17.7	17.4	18.4	19.1	19.5	18.6	14.6	10.9	9.5
		MCDBR	10.0	14.4	21.3	25.6	24.2	25.7	26.5	27.5	26.0	18.1	12.3	9.5
		MCWBR	8.9	10.8	13.2	14.5	13.1	12.5	11.0	11.5	12.8	11.0	9.6	8.6
	5% WB	MCDBR	9.6	12.4	17.5	22.4	21.8	22.6	23.0	23.4	21.4	15.8	11.2	9.3
MCWBR		8.9	10.3	11.8	13.3	12.5	12.2	10.4	10.9	11.7	10.5	9.3	8.7	
Clear-Sky Solar Irradiance	taub	0.321	0.340	0.371	0.393	0.399	0.402	0.396	0.387	0.366	0.356	0.330	0.315	
	taud	2.428	2.382	2.290	2.234	2.261	2.289	2.304	2.350	2.389	2.418	2.434	2.402	
	Ebn at Noon	236	257	266	271	274	273	272	270	265	247	229	220	
	Edh at Noon	21	27	35	41	41	40	39	36	31	26	21	19	
All-Sky Solar Radiation	RadAvg	306	558	914	1383	1630	1819	1732	1506	1179	685	374	261	
	RadStd	30	69	115	157	144	186	126	138	89	57	36	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page